Solution

Exercises: Differential Equation Descriptions for Systems

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1. A differential equation describes the output of a linear time-invariant system as (a weighted sum, a weighted product, a convolution) of (derivatives, differences, convolutions) of the output and the input.

2. The (coefficients, initial conditions, input) for the differential equation summarize all the past information about the system that is needed to determine its future.

3. The transient response of a system described by a differential equation is determined by the (roots of the characteristic equation, roots of a Fredholm equation, derivative of the input signal).

4. The steady-state response of an LTI system described by a differential equation (takes the same form as the input, depends on initial conditions, gives the impulse response).

5. Which of the following correspond to linear, constant coefficient differential equations? Select all that apply.

a)

$$y(t) + 2\frac{d}{dt}y(t) + 4\frac{d^2}{dt^2}y(t) = x(t) - \frac{d}{dt}x(t)$$

b)

$$y(t) + 2\frac{d}{dt}y^2(t) = x^2(t) - \frac{d}{dt}x(t)$$ ← $y^2(t)$ nonlin.

c)

$$y(t) = -2\frac{d}{dt}y(t) - 4\frac{d^2}{dt^2}y(t) + x(t) - \frac{d}{dt}x(t)$$

d)

$$y(t) + 2t^3\frac{d}{dt}y(t) + 3\frac{d^2}{dt^2}y(t) = x(t) - \frac{d}{dt}x(t)$$

time-varying coefficient

6. A system is described by the differential equation

$$\frac{d^2}{dt^2}y(t) + 5\frac{d}{dt}y(t) + 4y(t) = x(t) - \frac{d}{dt}x(t)$$

Find the characteristic equation and it's roots.

$$r^2 + 5r + 4 = 0$$

$$(r+4)(r+1) = 0$$

$$r_1 = -4$$

$$r_2 = -1$$

7. A system is described by the differential equation

$$\frac{d^2}{dt^2}y(t) + 5\frac{d}{dt}y(t) + 4y(t) = x(t) - \frac{d}{dt}x(t)$$

Recall that the transfer function is a ratio of polynomials, $H(s) = b(s)/a(s)$.

a) Find the coefficients of the numerator polynomial $b(s)$.

$b(s) = 1 - s$

b) Find the coefficients of the denominator polynomial $a(s)$.

$a(s) = s^2 + 5s + 4$

8. A system is described by the differential equation

$$\frac{d^2}{dt^2}y(t) + 4\frac{d}{dt}y(t) + 3y(t) = x(t) - \frac{d}{dt}x(t)$$

a) Suppose the input to this system is $x(t) = 2u(t)$. Find the steady-state response of the system. Hint: express the input for large t as $x(t) = \alpha e^{st}$.

$x(t) \longrightarrow 2e^{0t} \Rightarrow y_s(t) = 2H(0)e^{0t}, t \geq 0$

$H(s) = \dfrac{1-s}{s^2+4s+3}$ so $H(0) = \dfrac{1}{3}$

$y_s(t) = \dfrac{2}{3} \qquad t \geq 0$

b) Suppose the input to this system is $x(t) = e^{-4t}u(t)$. Find the steady-state response of the system.

$x(t) \rightarrow e^{-4t} \Rightarrow y_s(t) = H(-4)e^{-4t}, t \geq 0$

$H(-4) = \dfrac{1+4}{16-16+3} = \dfrac{5}{3}$

$y_s(t) = \dfrac{5}{3}e^{-4t}, t \geq 0$

9. A system is described by the differential equation

$$\frac{d^2}{dt^2}y(t) + 4\frac{d}{dt}y(t) + 3y(t) = x(t) - \frac{d}{dt}x(t)$$

Assume the system has initial conditions $y(t)|_{t=0^-} = 2$, $\frac{d}{dt}y(t)\big|_{t=0^-} = -4$ and that the input to the system is $x(t) = 0$.

a) Find the roots of the characteristic equation.

$r^2 + 4r + 3 = 0 \Rightarrow (r+3)(r+1) = 0$

$r_1 = -3$

$r_2 = -1$

b) Find the output of the system for $t > 0$. Note that since the input $x(t) = 0$, the output has zero steady-state component.

$y_t(t) = c_1 e^{-3t} + c_2 e^{-t}, \quad t > 0$

initial conditions translate to $t = 0^+$ since there are no impulses on the right hand side

$c_1 + c_2 = 2$

$-3c_1 - c_2 = -4$

$-2c_1 = -2$

so $c_1 = 1$

$c_2 = 1$

$y_t(t) = e^{-3t} + e^{-t}, \quad t > 0$

10. A system is described by the differential equation

$$\frac{d}{dt}y(t) + 3y(t) = 4x(t)$$

Assume the input is $x(t) = e^{-t}u(t)$ and the initial condition is $y(t)|_{t=0^-} = 1$. Find the output of the system. Identify the transient and steady-state components.

Steady-state component: $x(t) \to e^{-t}$

$y_s(t) = H(-1)e^{-t}$ $\quad H(s) = \frac{4}{s+3}$

So $y_s(t) = 2e^{-t}$, $t > 0$

Characteristic eqn: $r + 3 = 0 \Rightarrow r = -3$

transient response $y_t(t) = Ce^{-3t}$, $t > 0$

Output: $y(t) = Ce^{-3t} + 2e^{-t}$, $t > 0$

initial conditions at $t = 0^-$ translate to $t = 0^+$ $\quad y(0^+) = 1 = C + 2 \Rightarrow C = -1$

$y(t) = \underbrace{-e^{-3t}}_{\text{transient component}} + \underbrace{2e^{-t}}_{\text{steady state component}}$

11. A system is described by the differential equation

$$y(t) + 4\frac{d}{dt}y(t) + \rho\frac{d^2}{dt^2}y(t) + \gamma\frac{d^3}{dt^3}y(t) = x(t) - \frac{d}{dt}x(t) + 2\frac{d^2}{dt^2}x(t)$$

where ρ and γ are unknown constants. The input to the system is $x(t) = e^{-t}\cos(100\pi t)u(t)$. Of the following candidate steady-responses, which could NOT be correct? Select all that apply.

a) $y_s(t) = 2\sin(100\pi t)u(t)$ X no e^{-t}

b) $y_s(t) = 2e^{-t}\sin(100\pi t)u(t)$

c) $y_s(t) = e^{-t}u(t)$ X no $e^{j100\pi t}$

d) $y_s(t) = e^{-100\pi t}\sin(100\pi t)u(t)$ X needs e^{-t}, not $e^{-100\pi t}$

e) $y_s(t) = 2e^{-t}(\sin(100\pi t) + \cos(100\pi t))u(t)$

f) $y_s(t) = 100\pi t u(t)$ X no e^{-t} or $e^{-j100\pi t}$

input is of form $\frac{e^{-t}e^{j100\pi t}}{2} + \frac{e^{-t}e^{-j100\pi t}}{2}$

$= \frac{1}{2}e^{s_1 t} + \frac{1}{2}e^{s_2 t}$

$s_1 = -1 + j100\pi$, $s_2 = -1 - j100\pi$

Output (steady-state) must be in the same form

$y_s(t) = \frac{1}{2}H(s_1)e^{s_1 t} + \frac{1}{2}H(s_2)e^{s_2 t}$

13. Which of the following systems described by differential equations is likely to have the fastest transient response?

a)

$$\frac{d}{dt}y(t) + 3y(t) = 4x(t)$$

b)

$$\frac{d}{dt}y(t) + 300y(t) = x(t)$$

c)

$$\frac{d}{dt}y(t) - 300y(t) = 10x(t)$$

d)

$$\frac{d}{dt}y(t) + 0.5y(t) = 0.5x(t)$$

a) $r+3=0 \Rightarrow y_t(t) = ce^{-3t}$

b) $r+300=0 \Rightarrow y_t(t) = ce^{-300t}$

c) $r-300=0 \Rightarrow y_t(t) = ce^{300t}$

d) $r+0.5=0 \Rightarrow y_t(t) = ce^{-0.5t}$

case b decays the fastest and thus is likely to have the fastest response